

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2170

Chatom 138 kV
December 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2170. The Interconnection Customer proposes a 335.6 MW solar photovoltaic facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on August 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 644 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,364,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WADSWORTH to ST. MARTINS 138 kV	ST LOUIS to NUCEAR FUELS 138 kV L	109.5	210	20.2
CEDARVILLE JUNCTION to DETOUR 138 kV	S BELLEVILLE to NICHOLS 138 kV	108.0	566	22.0
CEDARVILLE JCT to PAGE 138 kV Line	NICHOLS SUBSTATION to S BELEVILLE 138 kV	107.9	571	22.0
FRITO LAY to TIMBERVILLE 138 kV Line	CHATOM to PURDY 138 kV	105.8	398	23.9
TIMBERVILLE to NEW SALISBURY 138 kV Line	PURDY to CHATOM 138 kV	105.5	404	23.9
SILOAM SPRINGS to DETOUR 138 kV	FT BENTON to DETOUR 138 kV Line	105.1	352	14.7
NORTHPOINT to S BELLEVILLE 138 kV	PURDY to PAGE SUBSTATION 138 kV	104.7	532	20.3
NORTHPOINT to MITCHELL GARDENS 138 kV	PURDY to PAGE SUBSTATION 138 kV Line	104.5	513	20.3
MITCHELL GARDES to KAPSTONE 138 kV	PAGE SUBSTATION to PURDY 138 kV L	104.5	501	20.3
RED TAIL to PLANT CITY 138 kV	SPRINGFIELD POWER LLC to EDINBURG 138 kV	100.4	246	9.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WADSWORTH to ST. MARTINS 138 kV	Reconductor line	\$1,364,000	33% (\$456,000)
CEDARVILLE JUNCTION to DETOUR 138 kV	Reconductor line	\$3,369,000	27% (\$898,000)
CEDARVILLE JCT to PAGE 138 kV Line	Reconductor line	\$2,742,000	27% (\$731,000)
FRITO LAY to TIMBERVILLE 138 kV Line	Reconductor line	\$1,568,000	20% (\$311,000)
NEW SALISBURY SUBSTATION to TIMBERVILLE 138 kV	Reconductor line	\$2,617,000	20% (\$520,000)
SILOAM SPRINGS to DETOUR 138 kV	Reconductor line	\$5,943,000	27% (\$1,599,000)
NORTHPOINT to S BELLEVILLE 138 kV	Reconductor line	\$2,370,000	27% (\$641,000)
NORTHPOINT to MITCHELL GARDENS 138 kV	Reconductor line	\$1,974,000	27% (\$534,000)
MITCHELL GARDES to KAPSTONE 138 kV	Reconductor line	\$1,538,000	27% (\$416,000)
PLANT CITY SUBSTATION to RED TAIL 138 kV	Reconductor line	\$1,714,000	15% (\$258,000)
	Total allocated to project		\$6,364,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.